

Lesson 06 · Power, constant, sum and difference rules

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Differentiate $y = 5x^2$.

☐ A $5x$

☐ B $10x^2$

☐ C $10x$

☐ D x^2

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Q2 · Rank 1 / Foundation

Differentiate $y = 2x + 3$.

☐ A $2x$

☐ B 5

☐ C 3

☐ D 2

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Q3 · Rank 1 / Foundation

Differentiate $y = ax + b$, treating a and b as constants.

☐ A a

☐ B b

☐ C $a + b$

☐ D ax

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Q4 · Rank 2 / Core

Differentiate $y = x^2 + x + 1$.

☐ A $2x + 2$

☐ B $2x + 1$

☐ C $x + 1$

☐ D $2x$

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Q5 · Rank 2 / Core

Differentiate $y = x^2 - 3x + 2$.

☐ A $2x + 3$

☐ B $2x - 1$

☐ C $2x - 3$

☐ D $x^2 - 3$

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Q6 · Rank 2 / Core

Differentiate $y = 2x^3 - 6x^2 + 49x - 54$.

☐ A $6x^2 - 6x + 49$

☐ B $6x^2 - 12x - 5$

☐ C $2x^2 - 6x + 49$

☐ D $6x^2 - 12x + 49$

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Q7 · Rank 3 / Application

Evaluate $f'(x)$ for $f(x) = 3x^9$ at $x = 1$.

☐ A 27

☐ B 3

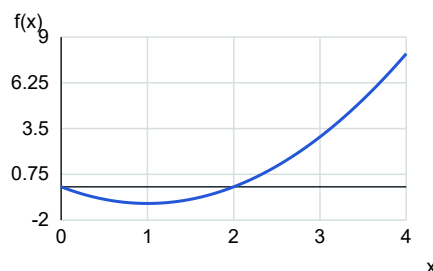
☐ C 9

☐ D 0

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Q8 · Rank 3 / Application

Evaluate $f'(x)$ for $f(x) = x^2 - 2x$ at $x = 3$.



☐ A 3

☐ B 4

☐ C 6

☐ D -4

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Q9 · Rank 3 / Application

Evaluate $f'(x)$ for $f(x) = x^3 - 4x^2 + 2x - 8$ at $x = 0$.

☐ A -8

☐ B 0

☐ C 2

☐ D -6

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Q10 · Rank 4 / Challenge · OPTIONAL

Evaluate $f'(x)$ for $f(x) = \sqrt{x} - \frac{2}{x}$ at $x = 4$.

☐ A $\frac{1}{8}$

☐ B $\frac{1}{2}$

☐ C $\frac{3}{2}$

☐ D $\frac{3}{8}$

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